

ABDULLAH AL-MAMUN

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Institution: Purdue University

RESEARCH INTERESTS

Database Systems, ML for Systems, Systems for ML

EDUCATION

Purdue University, USA

Expected: May 2027

Ph.D. in Computer Science

Thesis: Instance-Optimized Learned Indexes for Multi- and High-Dimensional Spaces

Advisor: *Walid G. Aref*

Members of the Advisory Committee: Elisa Bertino, Chris Clifton, and Jianguo Wang

Memorial University of Newfoundland, Canada

M.Sc. in Computer Science

Thesis: Anomaly Detection from Time-Changing Environmental Sensor Data Streams

Advisor: *Antonina Kolokolova*

Islamic University of Technology, Bangladesh

B.Sc. in Computer Science

Thesis: Mining Data Streams from a Higher Level of Abstraction

Advisors: *Mohammad Rezwanul Huq* and *M.A. Mottalib*

RESEARCH EXPERIENCE

Purdue University

West Lafayette, IN, USA

- **ML for Systems:** Applying machine learning techniques to build instance-optimized learned database systems components. For example, augmenting traditional multi-dimensional (e.g., R-tree) indexes with ML models and generating workload-specific filtered vector search accelerators for efficient query processing.

Memorial University of Newfoundland

St. John's, NL, Canada

- **Anomaly Detection:** Combining Kernel Density Estimation with an external change detector to detect contextual anomalies from time-changing sensor data streams.

PUBLICATIONS

1. **Abdullah Al-Mamun**, Yeasir Rayhan, and Walid G. Aref. “iFVS: Towards Instance-Optimized Filtered Vector Search.” (arXiv/preprint), 2026.
2. **Abdullah Al-Mamun**, Hao Wu, Qiyang He, Jianguo Wang, and Walid G. Aref. “A Survey of Learned Indexes for the Multi-dimensional Space.” ACM Computing Surveys (CSUR), Volume 58, Issue 4, pp. 1-37, 2025.
3. **Abdullah Al-Mamun**, Jianguo Wang, and Walid G. Aref. “Learned Indexes From the One-dimensional to the Multi-dimensional Spaces: Challenges, Techniques, and Opportunities.” In Companion of the 2025 International Conference on Management of Data (SIGMOD Tutorial Track), pp. 788-796. 2025.
4. **Abdullah Al-Mamun**, Ch Md Rakin Haider, and Walid G. Aref. “Query Processing Tradeoffs over an ML-Enhanced R-tree.” In Proceedings of the 8th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery (GeoAI@SIGSPATIAL), pp. 174-184. 2025.

5. **Abdullah Al-Mamun**, Ch. Md. Rakin Haider, Jianguo Wang, and Walid G. Aref. “The AI+R-tree: An Instance-optimized R-tree.” In proceedings of the 23rd IEEE International Conference on Mobile Data Management (MDM), pp. 9-18. IEEE, 2022.
6. **Abdullah Al-Mamun**, Hao Wu, Walid G. Aref. “A Tutorial on Learned Multi-dimensional Indexes.” In proceedings of the 28th International Conference on Advances in Geographic Information Systems (SIGSPATIAL, Tutorial Track), pp. 1-4, 2020.
7. **Abdullah Al-Mamun**, Antonina Kolokolova, Dan Brake. “Detecting Contextual Anomalies from Time-Changing Sensor Data Streams.” In Proceedings of the Doctoral Consortium of 25th ECML PKDD, p.13, 2015.

TEACHING EXPERIENCE

Department of Computer Science, Purdue University
Graduate Teaching Fellow (Instructor)

West Lafayette, IN, USA
August 2022- May 2023

- CS 348: Information Systems

Department of Computer Science, Purdue University
Graduate Teaching Assistant

West Lafayette, IN, USA
August 2017- May 2025

- CS 448: Introduction to Relational Database Systems Implementation
- CS 348: Information Systems

AWARDS AND HONORS

Summer 2025: Microsoft Fellowship to attend the EDBT’25 Summer School on AI & Data Management.

Fall 2022-Spring 2023: Graduate Teaching Fellow, Department of Computer Science, Purdue University.

Spring 2023: Graduate Teaching Award, Department of Computer Science, Purdue University.

2018: Summer Research Grant, Purdue University.

2015: Fellow of the School of Graduate Studies, Memorial University of Newfoundland.